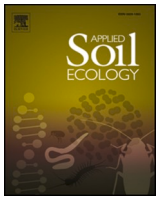
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Transforming morphology into ecology: Functional trait-based estimation of biomass and carbon budget in predatory nematodes (Nematoda: Mononchida)

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ABSTRACT

Predatory nematodes in the order Mononchida play a key biological regulation role in soil food webs but remain underexplored in terms of their ecosystem functions. Estimating biomass using morphological traits provides a practical method for assessing their contribution to carbon dynamics. This study calculates individual biomass and daily carbon budgets for mononchids using taxon-specific body-size data from publicly available literature. We calculated biomass and carbon budgets for 398 populations worldwide, representing 234 species, 47 genera, 12 subfamilies, and 8 families using Andrássy's formula. The estimated average biomass (fresh weight) was 6.71 μg for females and 10.75 μg for males across all species with available body size measurements, with corresponding daily carbon budgets of 0.07103 μg and 0.10481 μg, respectively. Compared to previous estimates, our data support higher biomass and more contribution of predatory nematodes to carbon budget. Biomass variability was notable across certain taxa (typically taxa within the family Mononchidae) but not all. Additionally, we tested the *G* index (female gonad-to-body length ratio) as a proxy for carbon budget estimation but surprisingly found no significant correlation between these two parameters ($r = 0.174$, $p = 0.087$). This study provides a comprehensive dataset for biomass and carbon budget estimation in predatory nematodes, enhancing understanding of their functional roles in carbon dynamics and ecosystem processes.

1. Introduction

Trait-based approaches are increasingly used for mechanistic understanding of the ecological strategies of soil invertebrates and their contributions to ecosystem functions (de Bello et al., 2021; Joimel et al., 2024) including organic matter dynamics (Bonfanti et al., 2024). Such trait-based studies have also extended to nematodes (Mulder and Maas, 2017; Zhang et al., 2024a, 2024b). For instance, the "nematode economics spectrum" (NES) framework (Zhang et al., 2024a) categorizes key traits, such as body length, width, length-width ratio, and biomass, under morphological traits. However, the link between these traits and ecosystem functions, including carbon dynamics, remains underexplored particularly for the least frequent nematode trophic group namely predatory nematodes which their ecological functions are understudied.

Carbon serves as a foundation of ecosystem processes (du Preez et al., 2022; Kopittke et al., 2022). Organic matter derived from living and dead organisms is an essential resource for decomposers and detritivores, with higher trophic level soil fauna regulating C dynamics through interactions with their prey (Angst et al., 2024). Quantifying the contributions of soil biota to C dynamics is therefore vital to understand ecosystem functioning. Nematodes are cosmopolitan organisms and are by far the most abundant animals on Earth (Bardgett and van der Putten, 2014) which their biomass is nearly equals to that of humans on the planet (van den Hoogen et al., 2019). Therefore, they play pivotal roles in organic matter transformations within and across landscapes (Ferris, 2010) by translocation or grazing of microbial populations (Angst et al., 2024). Predatory nematodes as a part of the nematode community increase soil food web structure and complexity by regulating biological population and facilitating energy turnover across trophic levels (Tu et al., 2024), indirectly contributing to carbon and other nutrients transformations in soil ecosystems (Whalen et al., 2013).

In ecological and biological contexts, a carbon budget refers to the balance of carbon inputs and outputs within an organism, population, or

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ecosystem over a specific period (Chapin et al., 2011). Nematode biomass, estimated from body size traits, provides a mean to assess their carbon budget, bridging morpho-taxonomy and ecosystem functions (Ferris, 2010; van den Hoogen et al., 2019). While biomass has already been estimated for trophic groups and families across different feeding habits and c-p groups (Ferris, 2010; van den Hoogen et al., 2019), more comprehensive datasets are needed to capture inter- and intra-taxonomic variability. Such studies can enhance our understanding of nematode ecology in soil ecosystems particularly with knowing that biomass-based studies can broaden our knowledge on the ecology of nematodes in soil ecosystems (Elbrecht and Leese, 2015; Schenk et al., 2019; Zhang et al., 2024b). Nematode biomass is typically calculated from body length (L) and greatest diameter (D) using Andrássy's formula (Andrássy, 1956; van den Hoogen et al., 2019; Zhang et al., 2024b) but can be estimated using body diameter (D) alone (Zhao et al., 2019) when length is not available.

The relationships between morphological traits and ecosystem functions are not limited only to body size traits. According to the theory of dynamic energy budgets (Kooijman, 2000; Jager et al., 2005), the carbon assimilated by an organism (e.g., a nematode) from its food is partitioned into somatic or maturity reserves in a fixed fraction (k). The partitioned carbon ultimately contributes to growth or reproduction while some is excreted as faeces or lost through respiration (Jager et al., 2005). Since reproductive output is an essential component of population dynamics, this connection suggests that morphological traits like reproductive system size may serve as indicators of carbon flow within an ecosystem, potentially linking individual physiology to broader ecological processes like nutrient cycling. Therefore, it can be inferred that the amount of carbon ingested by nematodes may be linked to the size of their reproductive system (Ferris and Benavides, 2024). Reproductive system in taxonomic literature is expressed as the direct length of nematode gonad or as *G* index which is the percentage of gonad length to the body length (Ye and Hunt, 2021).

Predatory nematodes of the order Mononchida are known as soil tigers (Ahmad and Jairajpuri, 2010) and inhabit a wide range of terrestrial and limnic biotopes worldwide (Andrássy, 2009). The Mononchida contain 454 known species and 45 valid genera (Andrássy, 2009). There are two suborders in Mononchida with most taxa belonging to Mononchina and the remaining taxa belonging to Bathyodontina. All nematodes in the order Mononchida have been categorized in group 4 according to c-p classification (Ferris, 2010). Members of the suborder Bathyodontina are found sporadically in soil samples with some found in aquatic habitats. They are considered as bacterivore nematodes (Andrássy, 2009; Ferris, 2010) but are traditionally classified under Mononchida. The suborder contains two superfamilies, Cryptonchoidea and Mononchuloidea, includes 4 genera and 12 species (Andrássy, 2009). Members of Mononchina are mostly soil dwellings organisms, but a smaller portion inhabit freshwater ecosystems. They are generally predators (carnivores) feeding on various types of soil and aquatic organisms such as protists, rotifers, oligochaetes and other nematodes (Andrássy, 2009). They possess a roomy and strongly sclerotized buccal cavity (stoma) which is armed with one or more large teeth and often also with small teeth. Their pharynx is a cylindrical tube with any morphologically distinct parts (Fig. 1). The suborder Mononchina

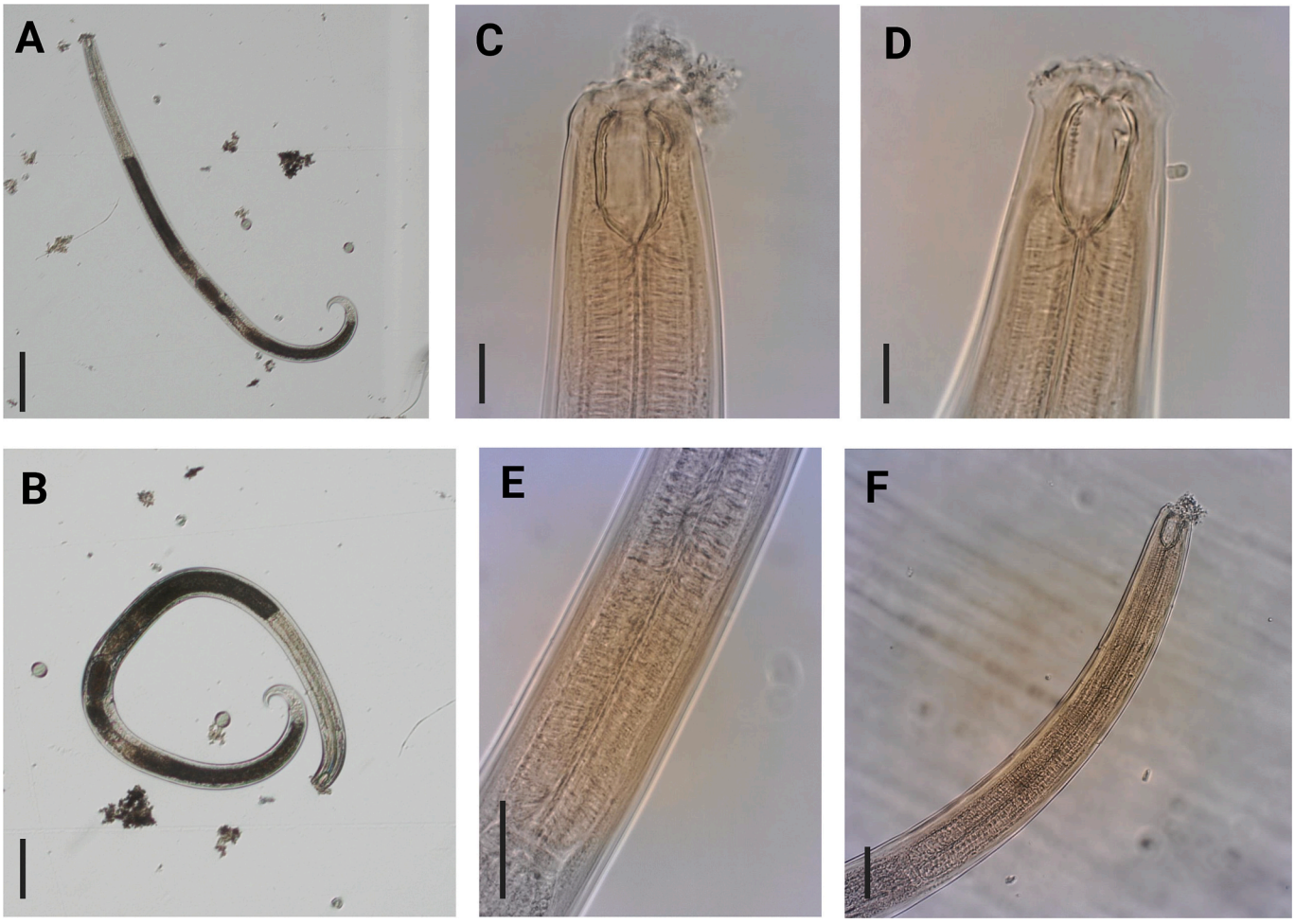


Fig. 1. Typical predatory nematodes of the order Mononchida. A, B: Entire body and general shape; C, D: Stoma or buccal cavity; E: Pharyngeal region; F: Anterior end showing stoma and pharyngeal region. Scale-bars: A, B = 100 μm; C, D = 10 μm; E, F = 20 μm.

includes two superfamilies, Mononchoidea and Anatonchoidea, and in total, contains 41 valid genera and 442 valid species (Andrássy, 2009).

Using Andrássy's (1956) formula, we estimate biomass and potential carbon budgets for predatory nematodes in the order Mononchida. Based on a database of taxon-specific body size traits from publicly available literature, we analyze 398 populations (nematode individuals collected from the same sample) worldwide, representing 234 species, 47 genera, 12 subfamilies, and 8 families. Our estimates provide individual biomass and daily carbon budgets across multiple taxonomic ranks and life stages. We also test the possibility of using *G* index as a proxy for estimating of carbon budget in nematodes. We hypothesize that there is a strong correlation between *G* index (as a morphological trait) and Pc (as an ecosystem function) which is a part of carbon assimilated in the nematode body. The biomass estimates will be a valuable resource to researchers in estimating soil biota contributions to carbon dynamics.

2. Material and methods

In this study, nematode biomass, expressed as fresh weight (µg), was calculated for all species using the classic formula (Andrássy, 1956): $W_{fresh} = LD^2/1.6 \times 10^6$, where W is the fresh weight (µg) per individual, L is the nematode body length (µm) and D is the greatest body diameter (µm). Since most studies report the average length of mature individuals and the length-to-diameter ratio (*a*) for their reported populations, *D* was estimated as $D = L/a$ when not explicitly provided. Dry weight (W_{dry}) was calculated as $0.104\ LD^2/1.6 \times 10^6$ following Mulder et al. (2005). The individual daily carbon budget was estimated assuming a constant temperature of 15 °C according to van den Hoogen et al. (2019). Daily production (*Pc*) and respiration (*R*) were calculated as $0.104 \times Wt/12\ cp_t$ and $0.0159 \times Wt^{0.75}$, respectively, where cp_t is the cp value of taxon t (Ferris, 2010; van den Hoogen et al., 2019). The total daily carbon requirement (*Ctot*) was determined as the sum of respiration and production (Ferris, 2010; van den Hoogen et al., 2019).

All predatory nematode indices were calculated using a database of taxon-specific body-size values (length and greatest body diameter) compiled from publicly available literature including reference books (Andrássy, 2009; Ahmad and Jairajpuri, 2010) and peer-reviewed papers published in international journals. Only studies were included that have identified the nematodes at species level. The average of the body size values was extracted based on the mentioned data in the study and not extrapolated from the figures. Calculations covered 398 populations worldwide, representing 234 species, 47 genera, 12 subfamilies, and 8 families (Supplementary Material).

Fresh and dry biomass, along with potential carbon budgets, were estimated separately for different life stages, including females and males (if present). Data on juveniles were scarce and available for only a few taxa. Additionally, *G* index values, representing the percentage of the female gonad to body length, were obtained from literature, and the Spearman correlation coefficient was calculated between *G* index and carbon used in reproduction (*Pc*) for each nematode population to assess the strength and direction of the monotonic relationship between the two variables. In total, data were available for 98 populations. Statistical analysis was performed using IBM SPSS Statistics version 29.0.0.

3. Results

3.1. Biomass and carbon budget estimation at higher taxonomic resolution: order to family rank

Taxonomic classification of Mononchida is summarised in Table 1. In the present work, the average of fresh and dry biomass and potential carbon budgets were calculated using Andrássy's (1956) model. We present and discuss data based mainly on fresh weight (fresh biomass; hereafter simply called biomass) but results on dry weight (dry biomass) are also presented in the Supplementary Material.

The average estimated biomass across the whole order was 6.71 and 10.75 µg for females and males, respectively. The average daily carbon budget for a single individual in the order Mononchida was 0.07103 and 0.10481 µg for females and males, respectively. Our results showed that

Table 1
Calculated biomass (fresh and dry weight) and carbon budget data for individuals of free-living nematodes in the order Dorylaimida (at family and higher ranks) using the formula developed by Andrássy (1956).

Taxa		Fresh W (µg)	Dry W (µg)	Pc (µg/day)	R (µg/day)	C_{tot} (µg/day)	Included data[a]
Order Mononchida	Female	6.71	0.70	0.01454	0.05649	0.07103	12 sf, 47 g, 234 s, 398 p
	Male	10.75	1.12	0.02329	0.08152	0.10481	10 sf, 27 g, 105 s, 152 p
Suborder Bathyodontina	Female	1.06	0.11	0.00231	0.01623	0.01853	3 sf, 4 g, 8 s, 11 p
	Male	1.74	0.18	0.00376	0.02390	0.02766	2 sf, 2 g, 3 s, 3 p
Superfamily Cryptonchoidea	Female	1.40	0.15	0.00304	0.01990	0.02294	2 sf, 2 g, 4 s, 5 p
	Male	1.74	0.18	0.00376	0.02390	0.02766	2 sf, 2 g, 3 s, 3 p
Family Bathyodontidae	Female	1.39	0.14	0.00301	0.02005	0.02305	1 sf, 1 g, 2 s, 3 p
	Male	1.75	0.18	0.00378	0.02394	0.02772	1 sf, 1 g, 2 s, 2 p
Family Cryptonchidae	Female	1.43	0.15	0.00309	0.01968	0.02277	1 sf, 1 g, 2 s, 2 p
	Male	1.71	0.18	0.00372	0.02383	0.02754	1 sf, 1 g, 1 s, 1 p
Superfamily Mononchuloidea	Female	0.78	0.08	0.00169	0.01317	0.01486	1 sf, 2 g, 4 s, 6 p
Family Mononchulidae	Male	–	–	–	–	–	–
Suborder Mononchina	Female	6.87	0.71	0.01488	0.05761	0.07248	9 sf, 43 g, 226 s, 387 p
	Male	10.93	1.14	0.02368	0.08268	0.10636	8 sf, 25 g, 102 s, 149 p
Superfamily Anatonchoidea	Female	7.55	0.79	0.01636	0.06830	0.08467	4 sf, 19 g, 77 s, 122 p
	Male	6.96	0.72	0.01507	0.06558	0.08065	4 sf, 13 g, 36 s, 60 p
Family Anatonchidae	Female	8.45	0.88	0.01830	0.07575	0.09405	2 sf, 8 g, 32 s, 64 p
	Male	6.88	0.72	0.01491	0.06469	0.07961	2 sf, 8 g, 19 s, 40 p
Family Iotonchidae	Female	6.56	0.68	0.01422	0.06009	0.07431	2 sf, 11 g, 45 s, 58 p
	Male	7.10	0.74	0.01539	0.06734	0.08273	2 sf, 5 g, 17 s, 20 p
Superfamily Mononchoidea	Female	6.56	0.68	0.01422	0.05286	0.06708	5 sf, 24 g, 149 s, 265 p
	Male	13.61	1.42	0.02949	0.09421	0.12370	4 sf, 12 g, 66 s, 89 p
Family Cobbonchidae	Female	7.33	0.76	0.01589	0.06323	0.07912	1 sf, 3 g, 8 s, 9 p
	Male	7.43	0.77	0.01610	0.06629	0.08240	1 sf, 2 g, 5 s, 6 p
Family Mononchidae	Female	12.05	1.25	0.02611	0.08778	0.11389	2 sf, 7 g, 83 s, 129 p
	Male	18.70	1.94	0.04051	0.12459	0.16511	2 sf, 6 g, 44 s, 61 p
Family Mylonchulidae	Female	1.34	0.14	0.00291	0.01929	0.02221	2 sf, 14 g, 57 s, 137 p
	Male	1.18	0.12	0.00256	0.01760	0.02016	1 sf, 4 g, 17 s, 22 p

[a] p = populations; s = species; g = genera; sf = subfamilies. Fresh W: fresh weight (biomass); Dry W: dry weight; Pc: daily carbon used in production; R: daily carbon used in respiration; Ctot: total daily carbon budget.

from this total carbon, 20.5 % (females) and 22.2 % (males) is allocated to production (*Pc*), while the remaining is released through respiration (Table 1).

Comparisons at suborder rank revealed that Mononchina have significantly higher biomass than Bathyodontina (6.87 and 10.93 vs. 1.06 and 1.74 µg for females and males, respectively), with females and males of Mononchina contributing 4-fold more to the carbon budget than Bathyodontina (Table 1). It should be noted that higher diversity in Mononchina may increase the chance of larger species in this suborder compared to Bathyodontina which shows a very small suborder with lower diversity.

In the suborder Bathyodontina, females in superfamily Cryptonchoidea showed higher biomass values compared to Mononchuloidea (1.40 vs. 0.78). In the suborder Mononchina, females in the two superfamilies Anatonchoidea and Mononchoidea showed comparable values (7.55 vs. 6.56 µg), while males of Mononchoidea represented by a much higher biomass (13.61 vs. 6.96 µg) (Table 1).

A considerable variability at family rank was observed only for superfamily Mononchoidea (Table 1; Fig. 2). Two families in Cryptonchoidea showed comparable body sizes (ranges from 1.39 to 1.75 µg) being slightly larger than Mononchuloidea (0.78 µg). Two families in Anatonchoidea also showed similar ranges to each other (6.56 to 8.45 µg), but in Mononchoidea, the family Mononchidae is typically represented by larger species (average of biomass equivalent to 12.05 and

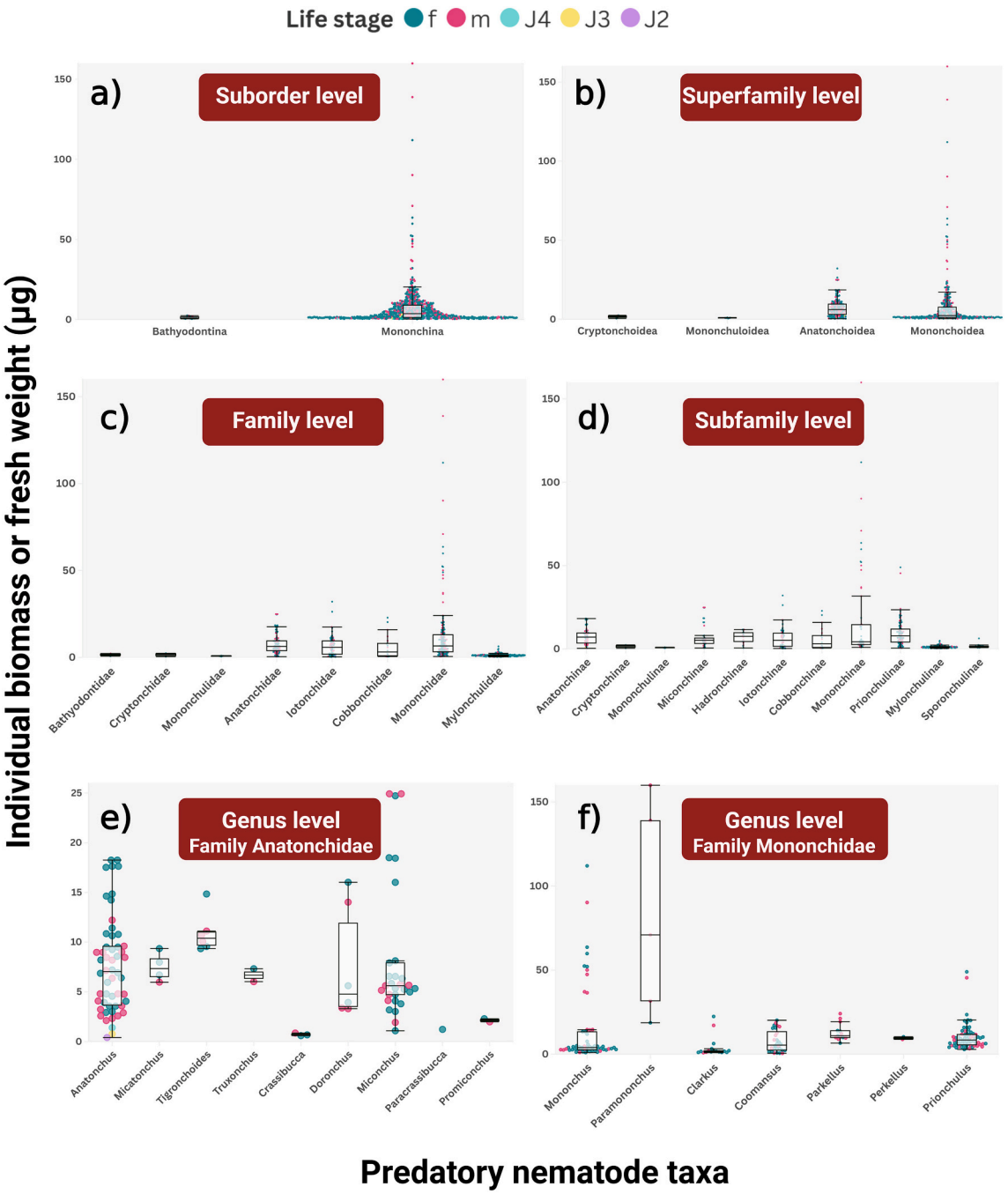


Fig. 2. Examples of Intra-taxon variability of individual fresh weight (biomass) for predatory nematodes. Order Mononchida at suborder (a), superfamily (b), family (c), subfamily (d), and genus (e, f) levels. All calculated by Andrássy (1956) formula. Nematode life stages include female (f), male (m), and the first- (J1), second- (J2), third- (J3) and fourth- (J4) stage juveniles.

18.70 μg for females and males, respectively) compared to smaller taxa in Cobbonchidae (7.33 and 7.43 μg for females and males, respectively) and Mylonchulidae (1.34 and 1.18 μg in females and males, respectively).

3.2. Biomass and carbon budget estimation at lower taxonomic resolution: subfamily to species rank

Great variability was observed for biomass within and among genera and subfamilies; for instance, from 2.9 to 18.2 μg (CV = 52) in *Anatonchus* (Anatonchinae), from 1.1 to 24.7 μg (CV = 76) in *Miconchus* (Miconchinae), from 0.4 to 32.1 (CV = 98) in *Iotonchus* (Iotonchinae), from 1 to 112 μg (CV = 188) in *Mononchus* (Mononchinae), from 1.0 to 22.4 μg (CV = 154) in *Clarkus* (Prionchulinae), from 0.6 to 20.2 μg (CV = 91) in *Coomansus* (Prionchulinae), and from 2.8 to 49.0 μg (CV = 69) in *Prionchulus* (Prionchulinae). However, some genera indicate comparably lower degrees of variation, such as *Mylonchulus* (Mylonchulidae) in which biomass ranges from 0.4 to 4.9 μg (CV = 57), with most species close to 1 μg. Most taxa show low rates of intra-variability at species rank, with maximum variations was observed for *Mononchus truncatus* (1.2–4.6 μg), and *M. aquaticus* (1.6–4.4 μg).

3.3. Biomass and carbon budget estimation across nematode life stages

Data on morphological traits of juvenile mononchids are notably scarce, likely because juvenile stages are generally considered less informative or diagnostically useful in traditional nematode taxonomy. As a result, most taxonomic descriptions focus on adult females and, when available, males, leaving juveniles underrepresented in morphometric datasets. This gap limits the precision of life-stage-specific ecological and functional assessments. In our dataset, female and male mononchids displayed broadly comparable biomass ranges across most families, with the exception of Mononchidae, where males exhibited markedly greater average biomass than females (18.70 vs. 12.05 μg). This suggests a potential case of sexual dimorphism in body size within this family, which may be linked to behavioural or reproductive roles that warrant further investigation.

Due to the limited availability of direct measurements for juvenile stages, we estimated biomass values using data derived from three well-described species namely *Anatonchus acutus*, *Iotonchus stockdilli*, and *Comiconchus trionchus* which had sufficient information for both adult and juvenile stages. Based on these species, we calculated female-to-juvenile biomass ratios, which revealed a consistent pattern of increasing biomass across developmental stages. Specifically, females were 2.5 times larger than J4 juveniles, 5.9 times larger than J3, 12.9 times larger than J2, and 29.0 times larger than J1 juveniles. Biomass ratios were not consistent between the three species as the ratios were smaller for *A. acutus* than the other two species.

3.4. Correlation between female gonad length and carbon used for production

In total, 98 populations represented G index estimates distributed in 66 species and 15 genera in the families Anatonchidae, Iotonchidae, Cobbonchidae, Mononchidae and Mylonchulidae. The results of G index estimates indicated an average of 14.3 for the order Mononchida but varied from 6.5 to 23.7 across taxa (Fig. 3). The correlation analysis revealed no significant relationship between G index and productive carbon allocation (Pc) estimates ($r = 0.174$; $p = 0.087$).

4. Discussion

Our estimates are generally comparable with earlier efforts based on body size of fewer species (Ferris, 2010), but show higher biomass for Mononchidae, Mylonchulidae, and Cobbonchidae by factors of 3.0, 2.2, and 1.8, respectively. Our dataset also includes a broader range of species, except for Bathyodonthidae (2 vs. 10 species in Ferris, 2010). Differences in taxonomic resolution can substantially influence the outcomes of biomass and carbon budget estimations. Studies utilizing broader taxonomic categories, such as family or order-level averages, may obscure significant intra-group variability and potentially misrepresent the ecological contributions of specific taxa. For instance, while

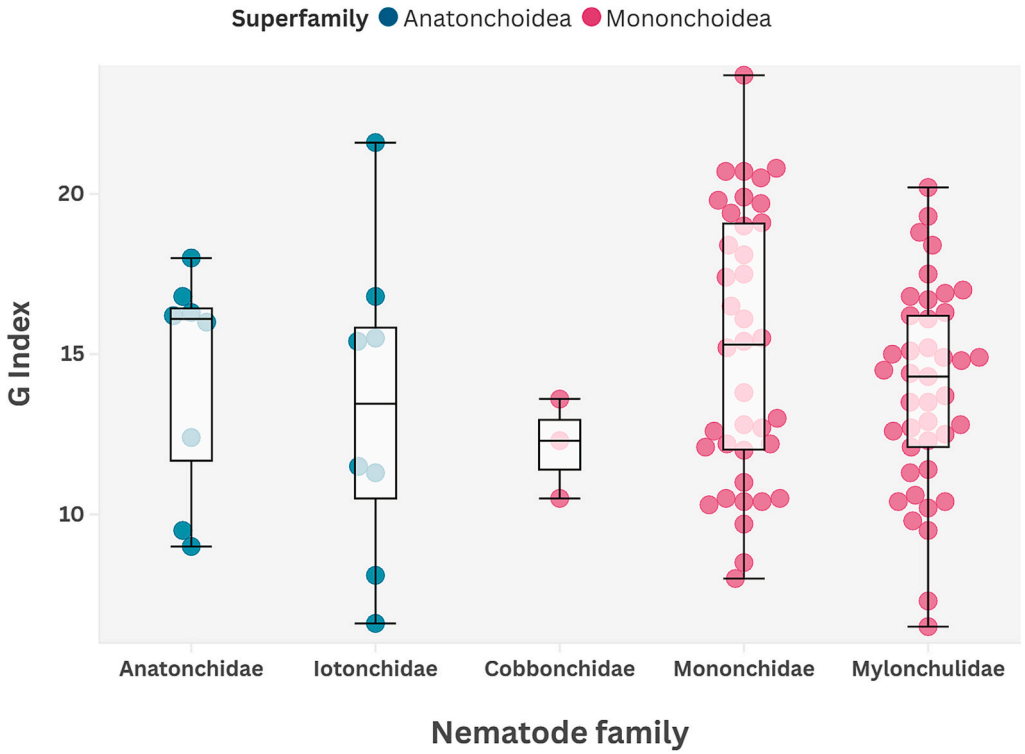


Fig. 3. Estimates of *G* Index (the percentage of the female gonad to the body length) for nematodes of certain families (with available data) in the order Mononchida.

our biomass estimates are generally consistent with earlier efforts based on limited species coverage (e.g., Ferris, 2010), our analysis based on a more extensive dataset encompassing 234 species across 47 genera revealed notably higher biomass values for Mononchidae, Mylonchulidae, and Cobbonchidae. These discrepancies likely stem from the inclusion of a wider diversity of species, which better captures the range of morphological and ecological variation within these families. Conversely, for families where our dataset is less comprehensive (e.g., Bathyodonthidae), biomass estimates may remain conservative or less reliable. This highlights the critical importance of fine taxonomic resolution in functional trait studies, as it enhances the precision of biomass assessments and improves the ecological interpretability of nematode contributions to soil processes such as carbon cycling.

Recent global estimates indicated that the total abundance of predatory nematodes is approximately 0.20×10^{20} individuals, less than that of other trophic groups including bacterivores, herbivores, fungivores, and omnivores (1.92×10^{20}, 1.25×10^{20}, 0.64×10^{20}, and 0.39×10^{20} individuals, respectively; van den Hoogen et al., 2019). Hence, their contribution to biomass (42.25 mt fresh weight) and monthly carbon budget (18.06 mt) is only higher than that of fungivores. The global study used average fresh biomass per individual predatory nematode of 2.11 μg, with a daily carbon budget of 0.0301 μg. Our updated estimates of predatory nematodes are considerably greater at 6.71 μg and 10.75 μg of fresh biomass, and 0.0710 μg and 0.1048 μg daily carbon budgets, for females and males, respectively. Notably, global estimates (van den Hoogen et al., 2019) are based on a juvenile-dominated community with a 7:3 juvenile-to-adult ratio. Additionally, the average biomass for Mononchida was higher than that of omnivores (Ghaderi et al., 2025) and herbivores (Ghaderi, 2020).

The pattern of biomass accumulation across life stages has important implications for ecological modelling, particularly in estimating community-level carbon budgets or energy fluxes in soil food webs. The adult:juvenile ratios demonstrate a steep growth trajectory throughout ontogeny, reflecting significant somatic investment during development. Ignoring juvenile stages or applying adult-based biomass values uniformly across all individuals may lead to considerable overestimations in community biomass and misrepresentations of trophic dynamics. Therefore, incorporating life-stage-specific biomass estimates, even if based on extrapolations from representative species, is crucial for improving the accuracy of ecological models and understanding the functional roles of predatory nematodes in soil ecosystems.

Globally, nematodes were estimated to contribute 0.14 Gt of carbon turnover during the growing season, with predatory nematodes accounting for 13 % (van den Hoogen et al., 2019). Our results show that only 20.5 % (females) and 22.2 % (males) of total carbon is allocated to production (*Pc*), while ~79 % is released through respiration, slightly lower than the 83.3 % global estimate (van den Hoogen et al., 2019). This difference may be due to the estimation of biomass and carbon budget at high taxonomic level (order) because using high taxonomic level averages can distort biomass calculations—overestimating when smaller taxa dominate and underestimating when larger ones prevail (Ferris and Benavides, 2024). This highlights the need for finer taxonomic resolution such as the framework was utilised in our study. No strong evidence for correlation was found between *Pc* and female reproductive gonad development, though incorporating more taxa in *G* index estimates or using mass-based methods may improve this relationship. Ferris and Benavides (2024) reported *G* index values exceeding 25 % for c-p4 nematodes, while our estimates ranged from 6.5 % to 23.7 %, typically between 10 % and 20 %. Extending such estimates across the entire nematode community with larger sample sizes could refine quantitative measures of c-p groups.

It is essential to recognize that the biomass and carbon budget estimates presented in this study are based on trait-based extrapolations and thus should be validated through direct empirical measurements. Controlled experiments using specific predatory nematode taxa or representative communities, whether in laboratory microcosms or in situ field conditions, are necessary to confirm the accuracy of these estimations. Such validation efforts would help refine the assumptions made in current models and provide a more reliable basis for incorporating nematode data into broader ecosystem-level carbon assessments.

A robust validation framework should also account for temporal dynamics in nematode abundance and community composition. Seasonal shifts, life stage transitions, and variations in extraction efficiency can all influence the representativeness of measured biomass data. These factors are particularly relevant for predatory nematodes, whose population dynamics and detectability may fluctuate significantly with environmental conditions. In addition, developmental stage plays a critical role, as life-stage-specific differences in size and metabolism can markedly affect biomass estimates. Including juvenile stages in validation experiments, as emphasized by Ghaderi (2020), can significantly enhance the ecological accuracy of nematode-based carbon models.

Furthermore, future research should address how key abiotic factors such as temperature, soil moisture, pH, and nutrient availability modulate nematode biomass and metabolic activity. Prior studies on bacterivorous (Procter, 1987) and freshwater nematodes (Maciute et al., 2021) have shown that metabolic rates and carbon allocation strategies vary under different environmental regimes. Applying similar experimental designs to predatory nematodes would improve our understanding of how external conditions influence their ecological roles. Integrating data on metabolic rates (Ferris, 2010) and life-stage-specific traits will ultimately lead to more refined models of carbon flow and nutrient cycling in terrestrial ecosystems, particularly when attempting to link soil biodiversity with functional outcomes.

CRediT authorship contribution statement

Reza Ghaderi: Visualization, Methodology, Investigation, Data curation, Conceptualization, Writing – review & editing, Writing – original draft. **Uffe N. Nielsen:** Validation, Methodology, Investigation, Conceptualization, Writing – review & editing. **Ji-Zheng He:** Validation, Supervision, Resources, Project administration, Funding acquisition, Conceptualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.apsoil.2025.106605.

Data availability

Data of this work are available online as Supplementary Data.

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